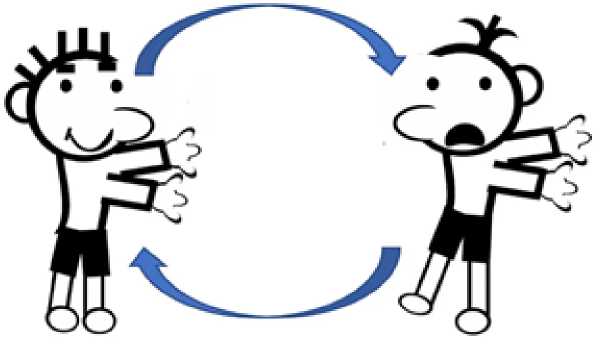
Boosting Bank Credit Card Marketing with **Uplift Modeling:** A Dataiku Approach

by Mohamad Ikhsan Nurulloh

Business Case

BankX conducted an A/B test on two email campaigns to encourage customers to apply for credit cards. Campaign A featured personalized messages with tailored offers, while Campaign B used generic messages. To understand the causal effect of these campaigns, BankX needs to isolate the impact of each campaign from other influencing factors. By analyzing the data, BankX aims to estimate the treatment effect of each campaign. This analysis will help refine marketing strategies, allocate resources more effectively, and explore new channels to enhance customer applications for credit cards.

Why Uplift Modeling



Purpose: Measures the incremental impact of marketing interventions, like email campaigns.

How It Works: Compares the additional effect of a treatment (e.g., personalized emails) vs. a control group (e.g., generic emails).

Application: Helps BankX assess which campaign drives more credit card applications.

Benefits: Optimizes marketing strategies, improves resource allocation, and enhances ROI.

Outcome: Better customer targeting and increased campaign effectiveness.

The Classic Uplift Segments

Will reach the outcome with or without treatment.

✗ Save time and money by not targeting.

Sure Things

The treatment will cause them to act. Truly incremental conversions.

✓ Target as many persuadables as possible.

Persuadables

Will **never** reach the outcome even with the treatment.

✗ Save time and money by not targeting.

Lost Causes

Will be less likely to purchase with treatment.

✗ Prevent negative effects.

Sleeping Dogs

Dataiku for Uplift Modeling

What is Dataiku: A platform for data science and machine learning with tools for preparation, modeling, and deployment.

Why Dataiku: Simplifies data cleaning, offers uplift modeling algorithms, supports team collaboration, and provides visualization tools.

Benefits for BankX: Streamlines uplift modeling, optimizes marketing strategies, and improves decision-making and resource allocation.

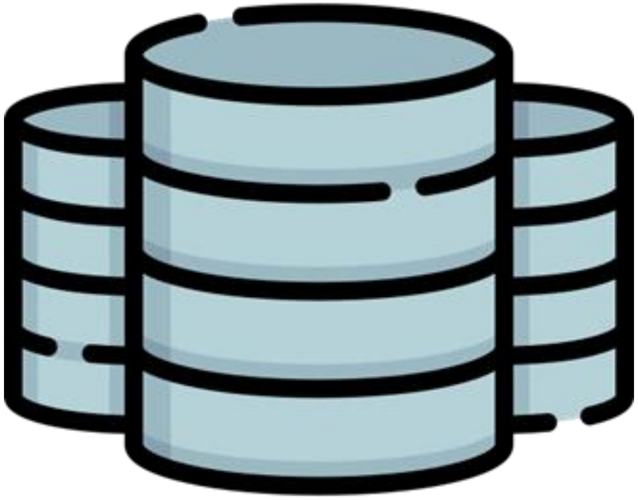
Dataset

Dataset

Dataset Summary:
- Records: 100,000 (synthetic data generated via Python)
- Format: CSV
- Purpose: Credit risk analysis, marketing, machine learning, statistical analysis

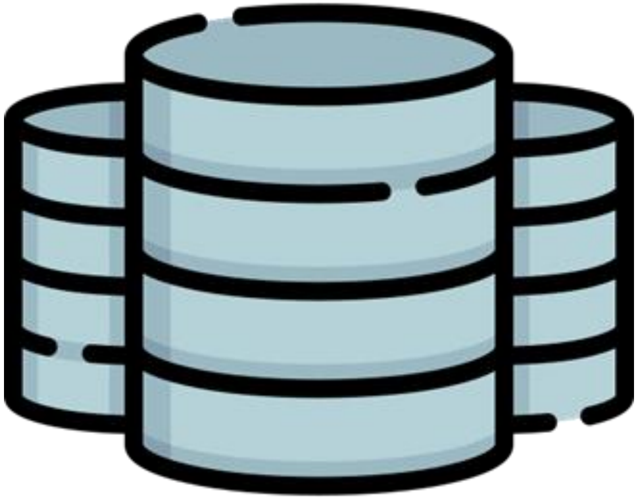
Key Attributes:
- Customer_ID: Unique identifier
- Demographics: Age, Gender, Education_Level, Employment_Status
- Financial: Income, Credit_Score, Average_Account_Balance, Loan_Amount, Loan_Term
- Banking Behavior: Average_Transactions_Per_Month, Credit_History
- Others: Occupation, Bank_Location

Distribution: Balanced distribution simulating realistic banking scenarios



https://github.com/ikhsannur1996/banking-data-dummy

Dataset



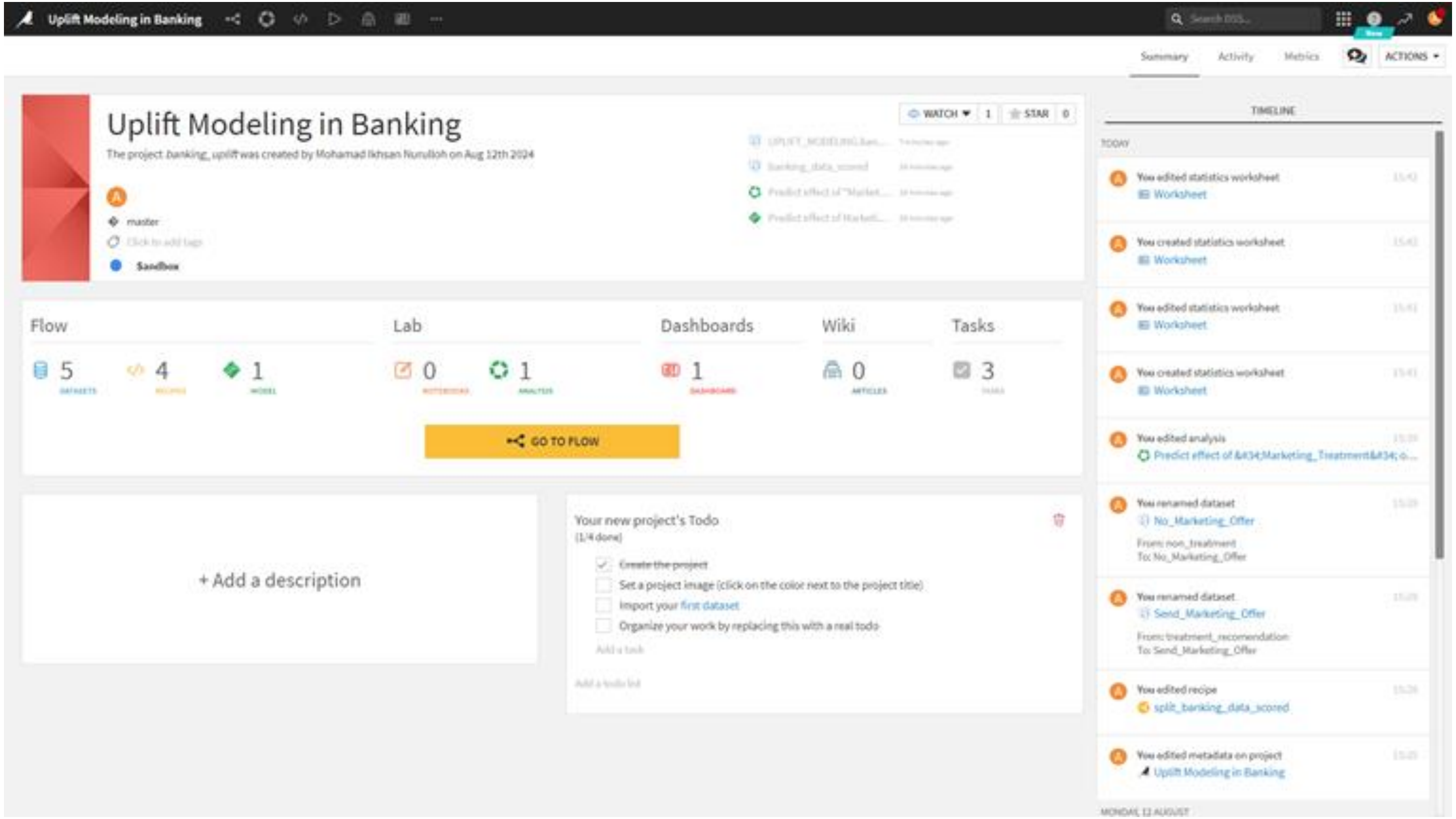
https://github.com/ikhsannur1996/banking-data-dummy

Disclaimer:

- This case study uses synthetic data generated via Python.
- The data may not be randomized, which could affect certain metrics.
- Metrics such as the Qini curve and AUUC assume randomized treatment.
- Non-randomized data may lead to unreliable results in these metrics.
- For accurate uplift modeling, it is recommended to use properly randomized real-world data.

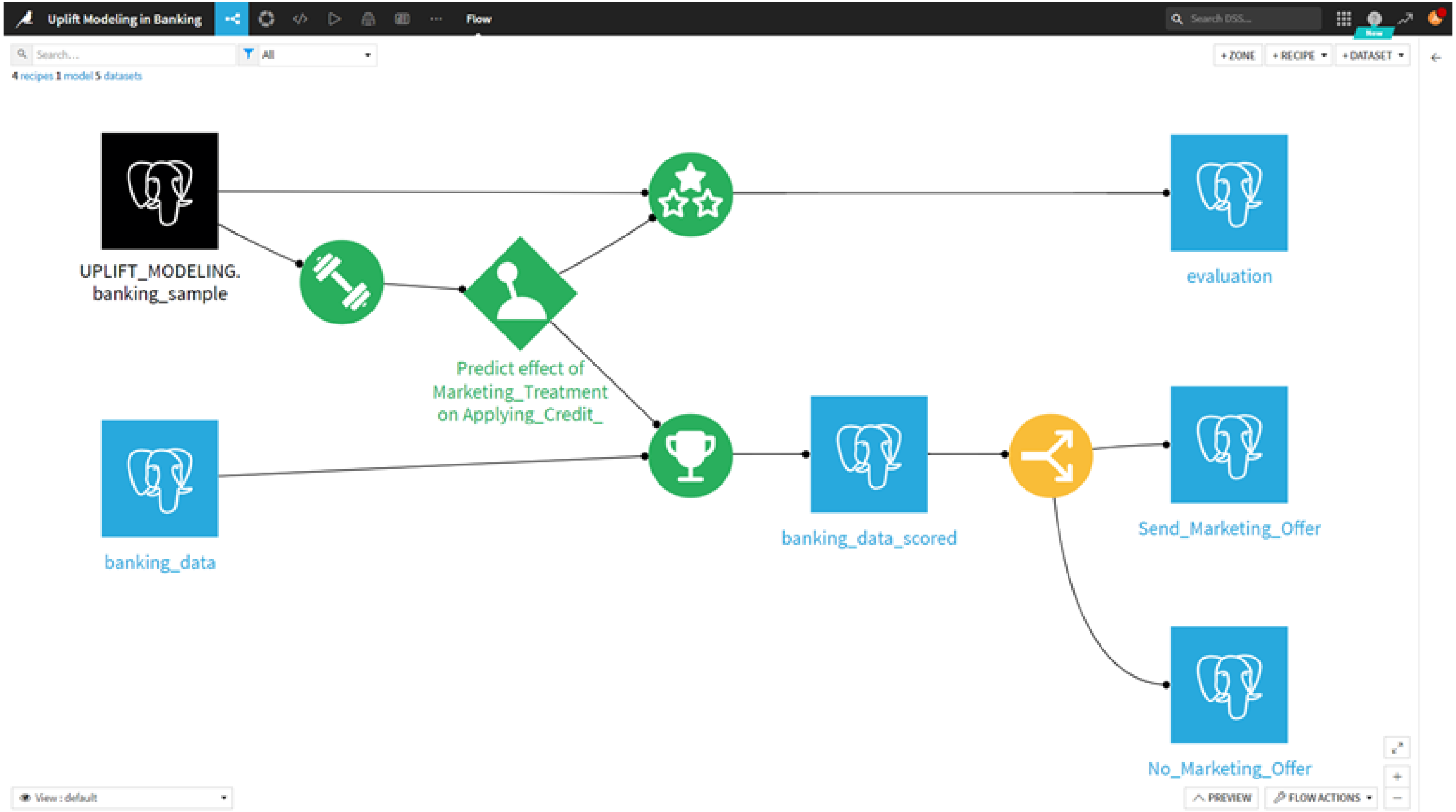
Dataiku Data Science Project for Uplift Modeling

Dataiku Uplift Modeling Project Page



The image displays a data science project in Dataiku with various tabs like "Flow," "Lab," and "Dashboards," along with sections that include navigation options, task progress bars, and action buttons.

Uplift Modeling Flow

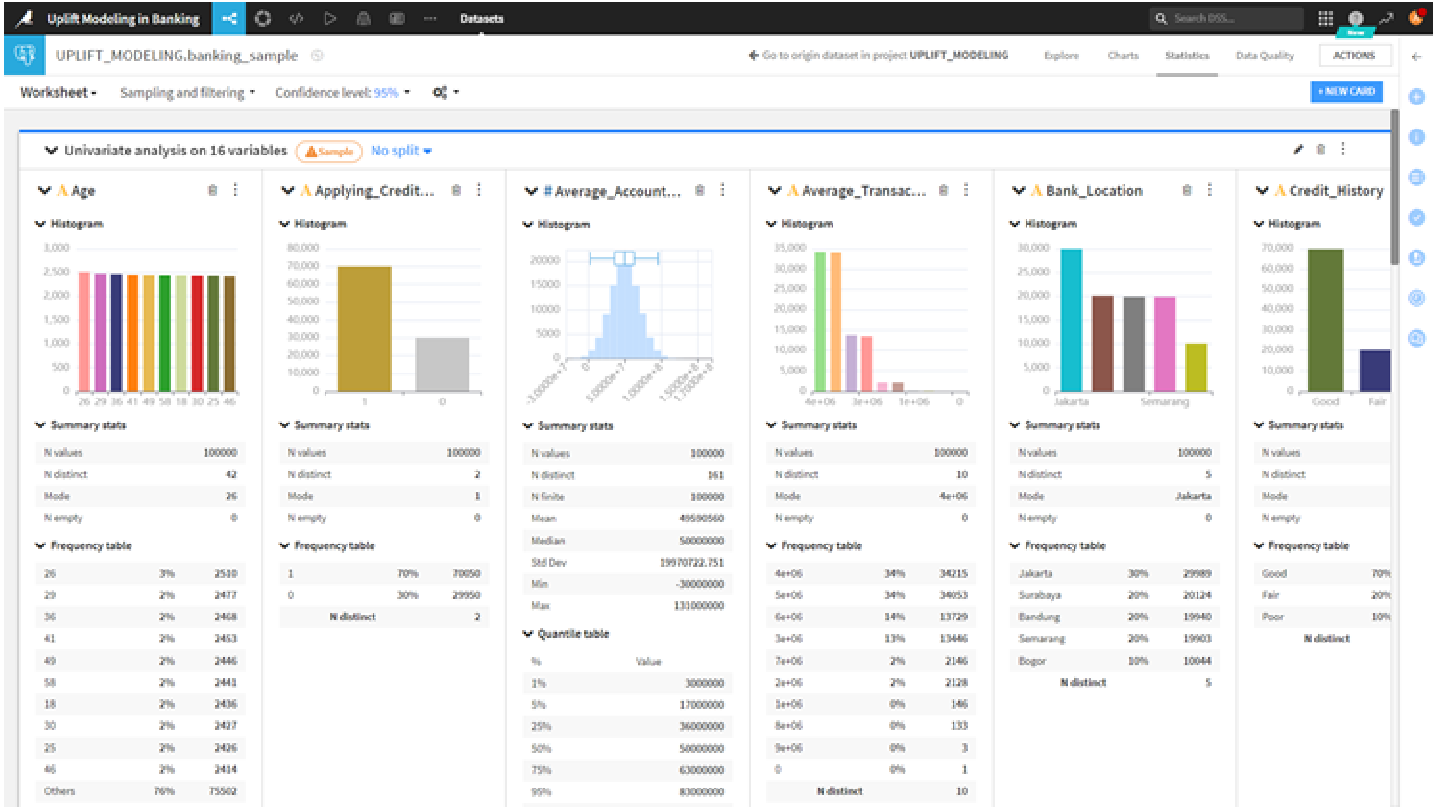


Uplift Modeling Flow Description

- **Overview:**
  - Purpose: Uplift modeling predicts the incremental impact of a treatment (e.g., marketing campaign) on customer behavior, useful for targeted marketing.
  - Platform: Dataiku, a data science platform.
- **Flowchart Breakdown:**
  - Inputs:
    - banking_modeling: Initial modeling data with customer features.
    - banking_data: Raw data from banking operations.
  - Decision Node:
    - Predict Effect: Model predicts the impact of the marketing treatment on the likelihood of customers applying for credit.
  - Paths:
    - Evaluation: Assess model predictions for accuracy using cross-validation and performance metrics.
    - Marketing Offers:
    - Send_Marketing_Offer: Positive uplift triggers sending offers via email, SMS, etc.
    - No_Marketing_Offer: No uplift or negative impact means no offer is sent, reducing costs.

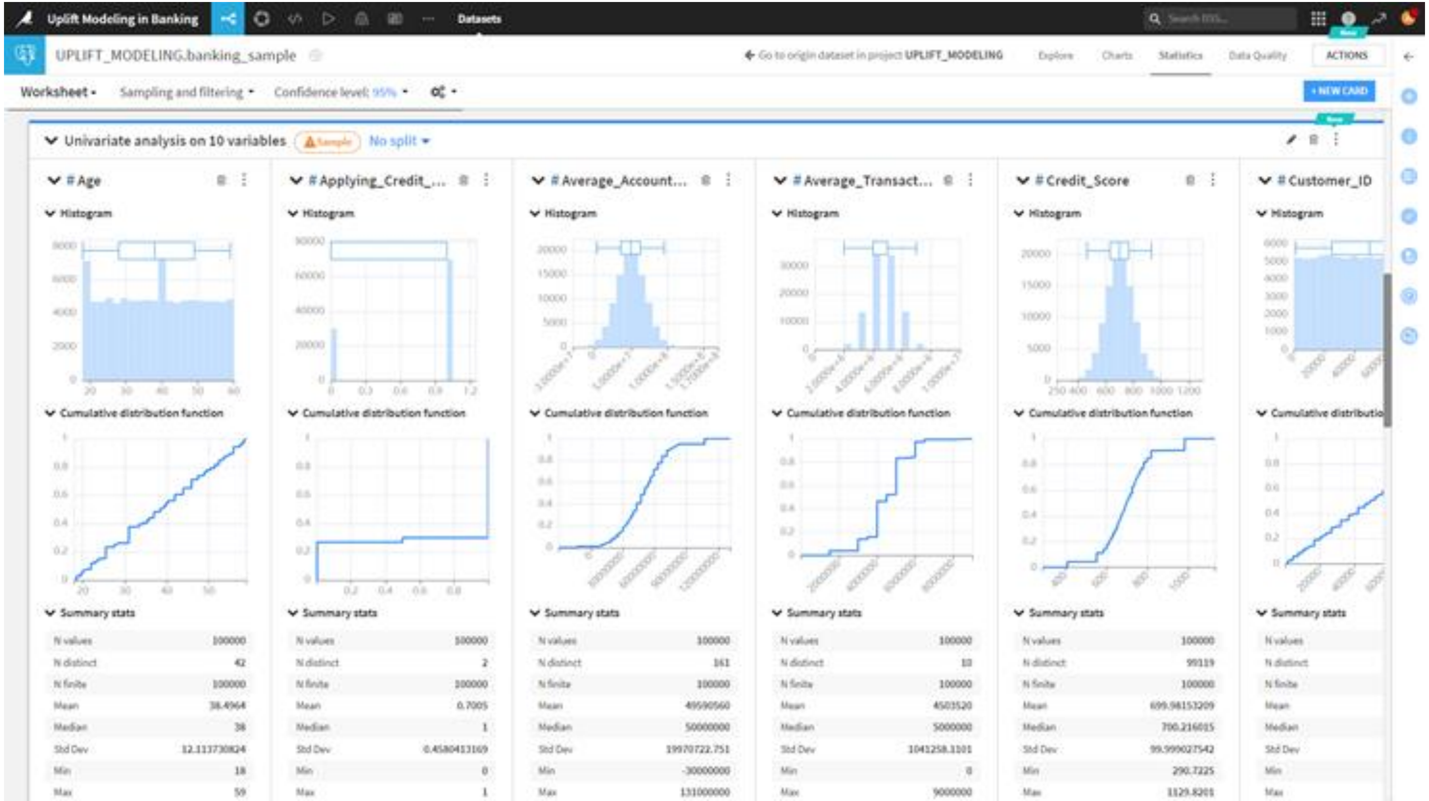
Exploratory Data Analysis with Dataiku

Exploratory Data Analysis with Dataiku



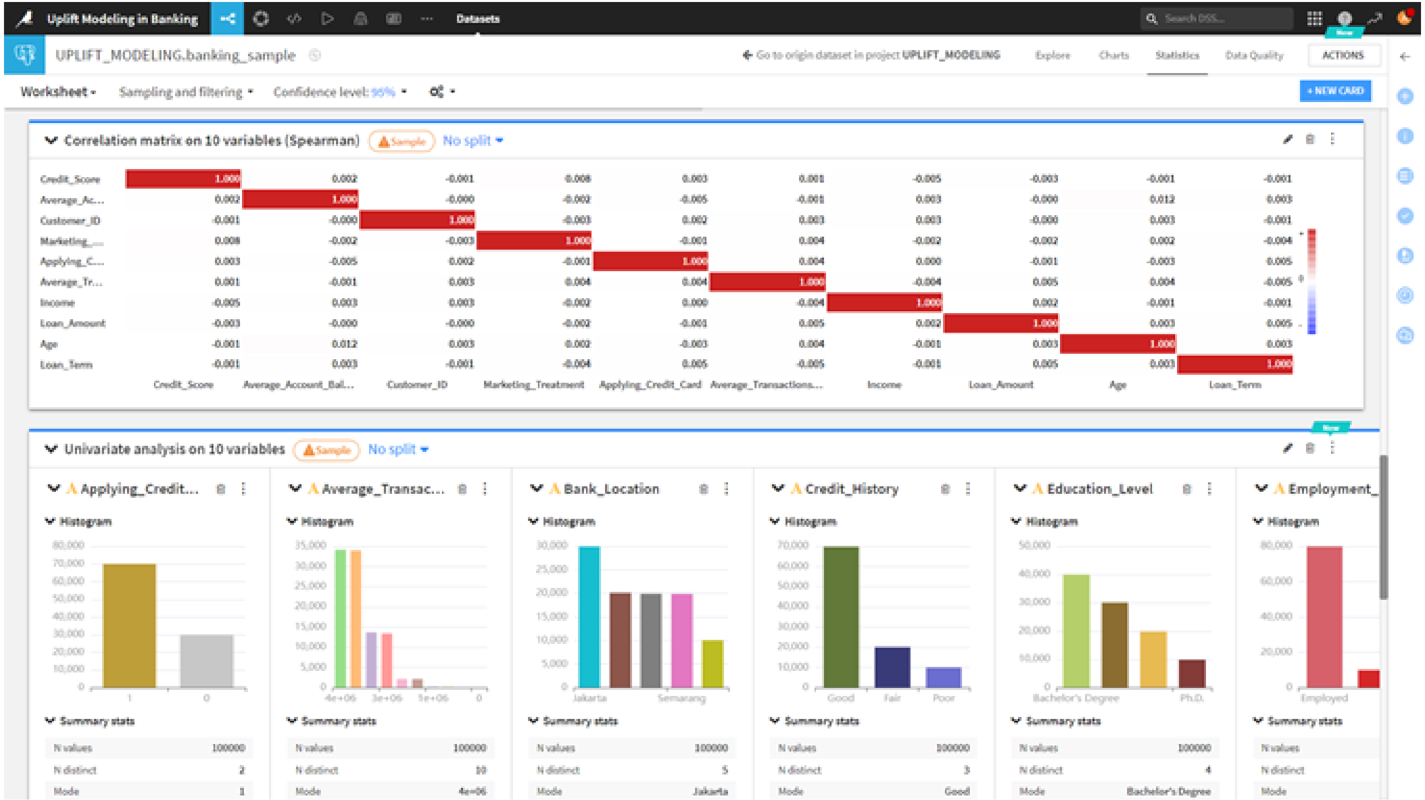
These charts represent banking data points. Dataiku offers features for generating statistics and visualizing data to analyze customer behavior and financial trends. This image is relevant for those in data science, statistics, or anyone analyzing large datasets to identify patterns and relationships.

Exploratory Data Analysis with Dataiku



These charts represent banking data points. Dataiku offers features for generating statistics and visualizing data to analyze customer behavior and financial trends. This image is relevant for those in data science, statistics, or anyone analyzing large datasets to identify patterns and relationships.

Exploratory Data Analysis with Dataiku



These charts represent banking data points. Dataiku offers features for generating statistics and visualizing data to analyze customer behavior and financial trends. This image is relevant for those in data science, statistics, or anyone analyzing large datasets to identify patterns and relationships.

Uplift Modeling Design with Dataiku

Uplift Model Outcome & Treatment Design

Outcome

Prediction type	Causal Classification
Outcome	Applying_Credit_Card
Preferred outcome class	1

Predicted effects will be computed with the predicted probabilities of this class

Treatment

Treatment variable	Marketing_Treatment
Control value	0

Treatment is considered binary: either equal to the control value (control group) or not (treated group)

Allow multi-valued treatment	☐
Drop missing values	☑

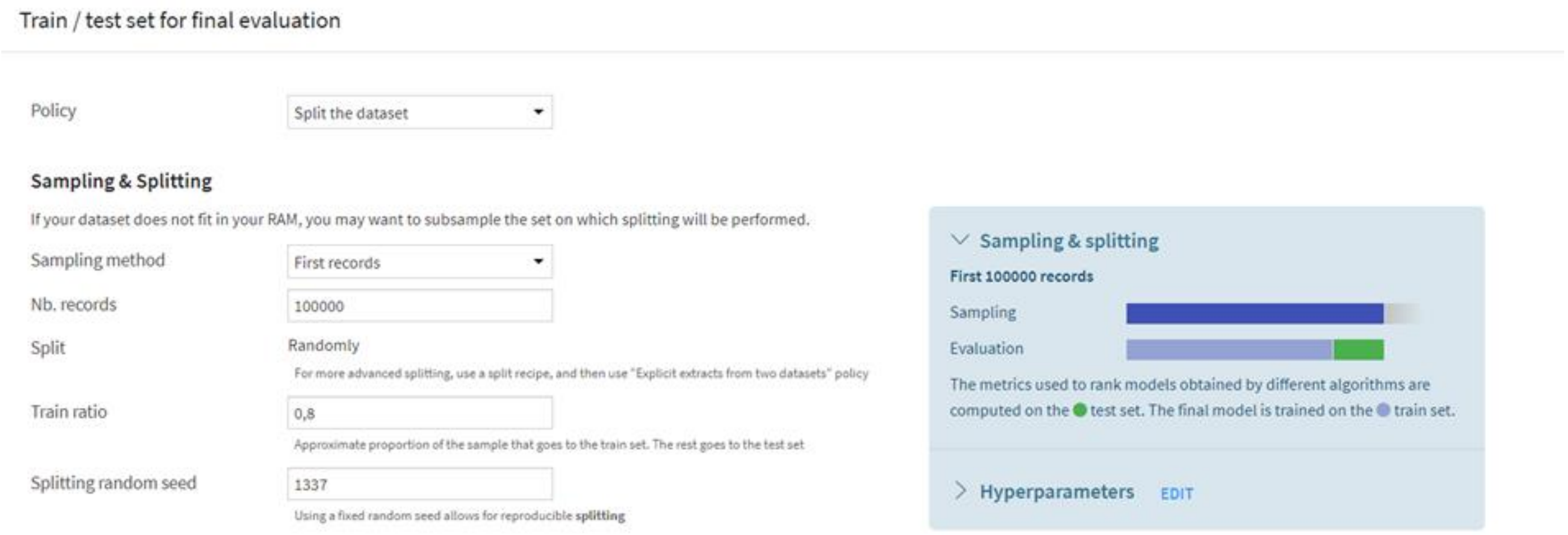
Estimated Average Treatment Effect

Assuming full randomization of the treatment variable (**Marketing_Treatm...**), the **Average Treatment Effect (ATE)** can be estimated by the difference of the frequency of the preferred class **Applying_Credit_C...** with treatment **Marketing_Treatm... is not 0** and without. The estimated ATE is `0.00041440` .

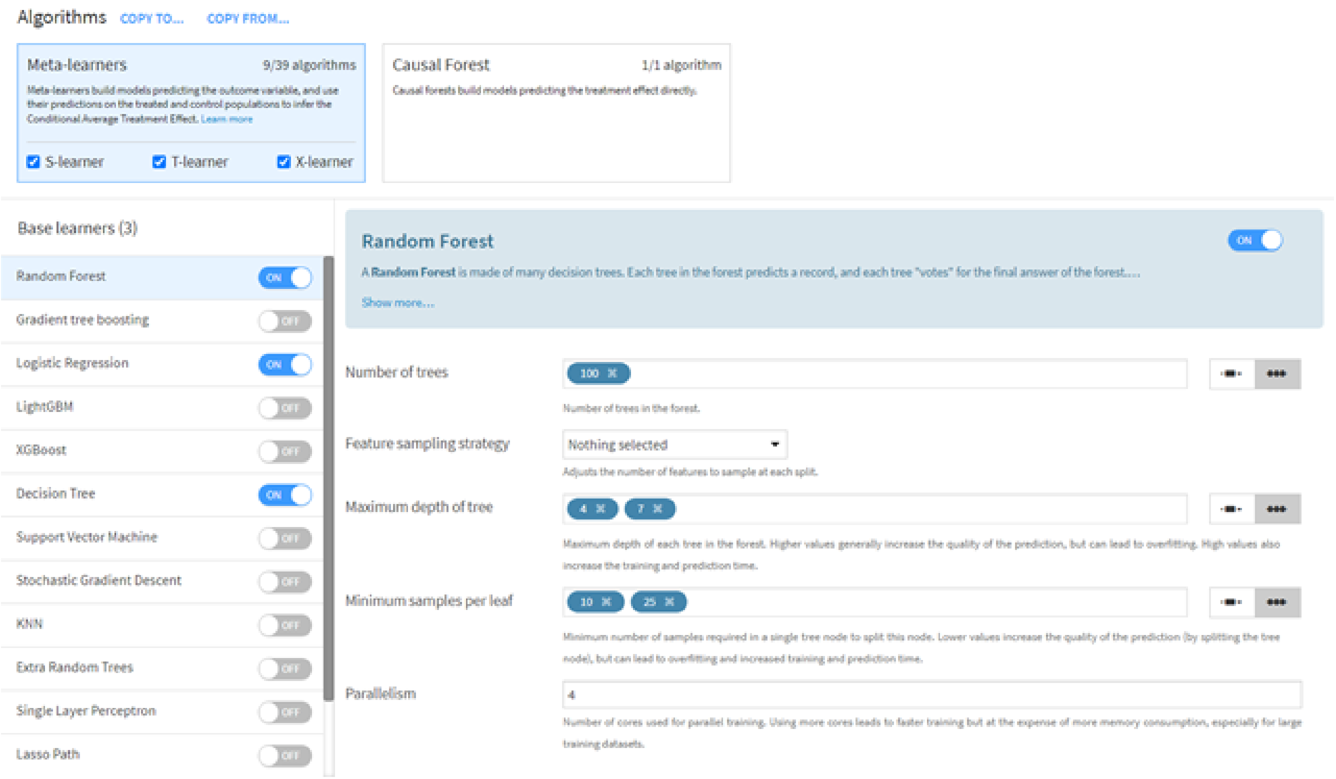
The values below are computed on a sample using the current sampling settings. You can edit them from the analysis script.

Treatment / Outcome value	0	1	All	Preferred class frequency
Control *Marketing_Treatment is 0*	1486	3524	5010	0.70339
Treatment *Marketing_Treatment is not 0*	1478	3512	4990	0.70381
All	2964	7036	10000	0.70360

Uplift Model Train/Test Set Design

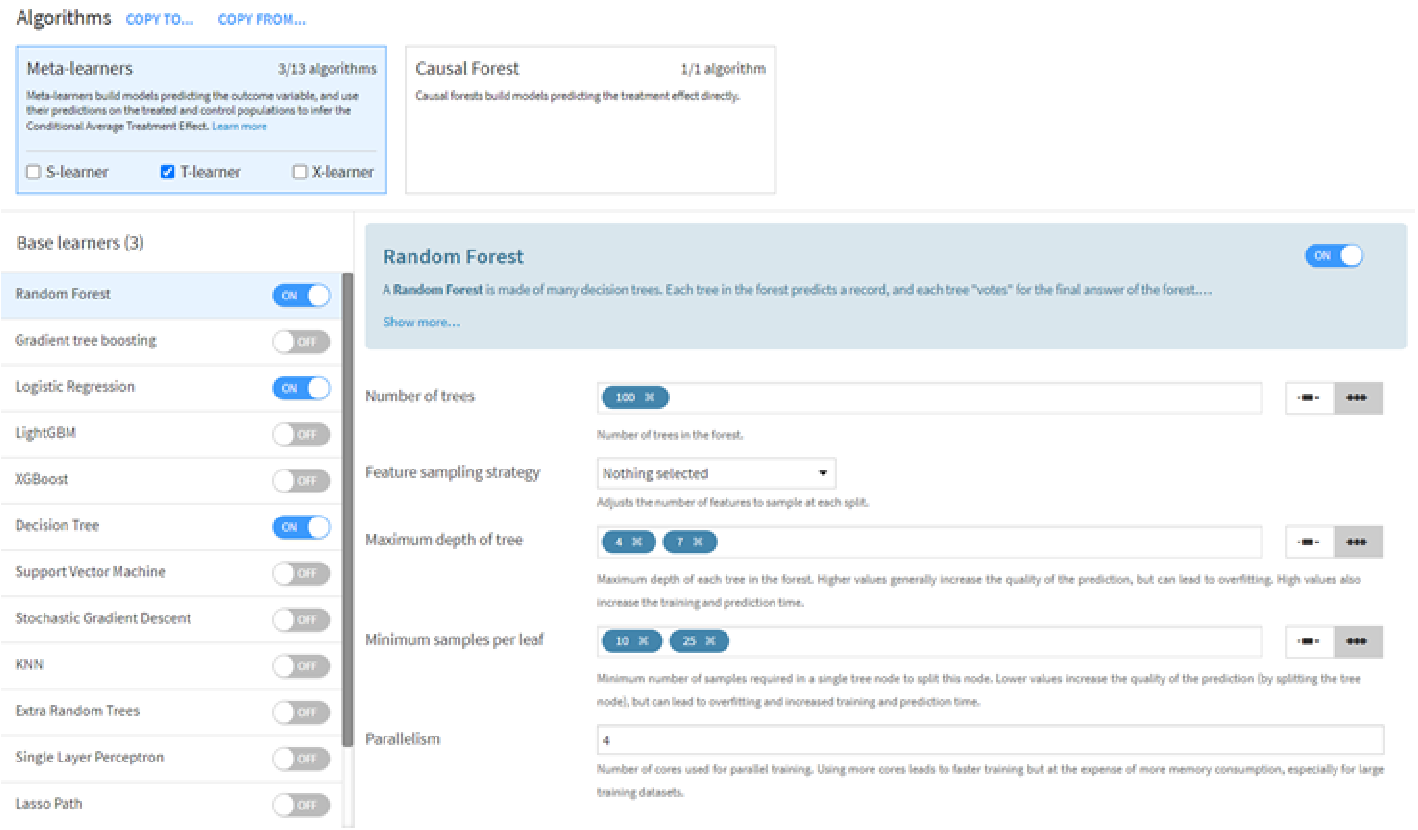


Uplift Modeling Algorithms



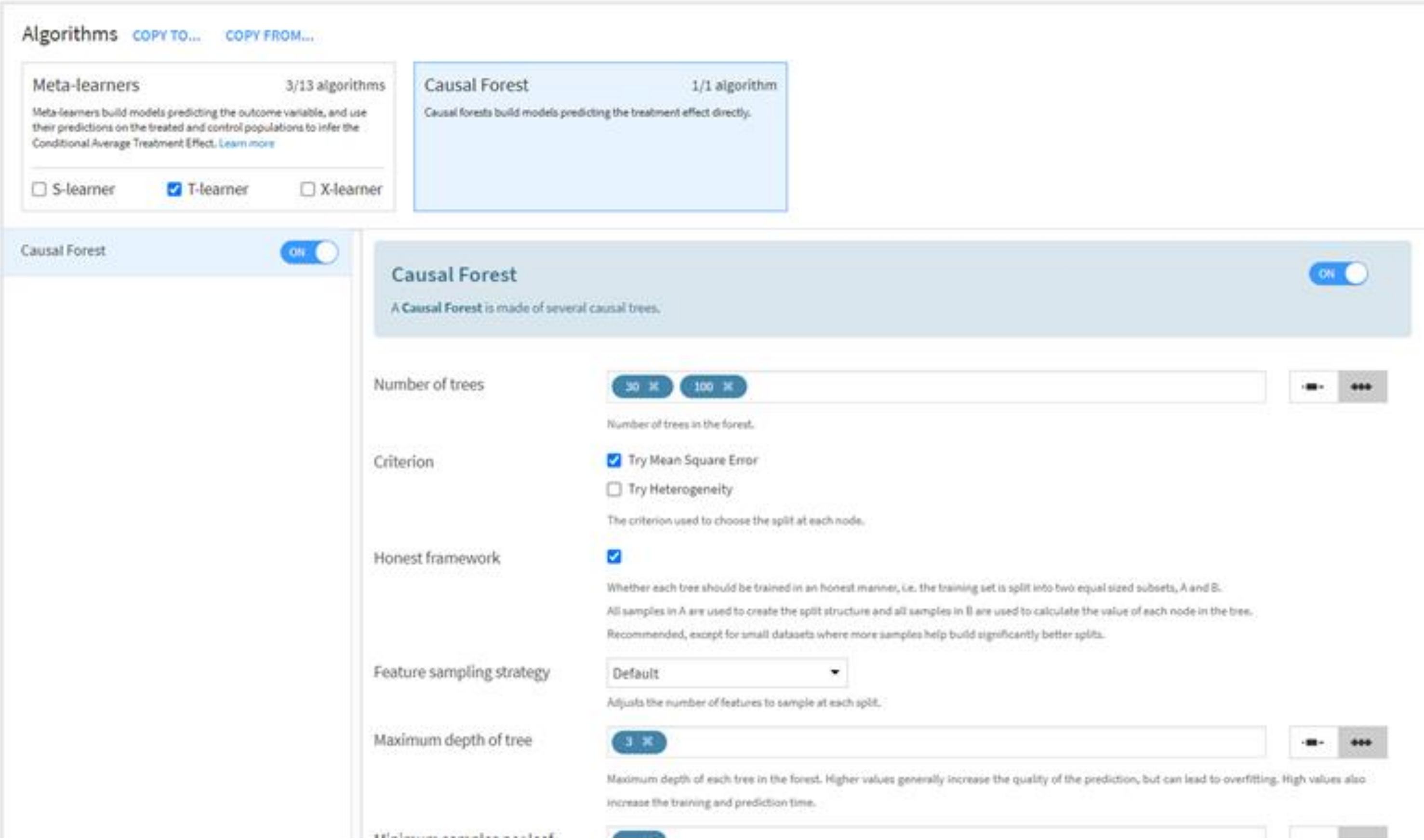
It shows dropdown menus, checkboxes, and input fields for different parameters like algorithms to use (e.g., Random Forest). Dataiku provides 4 groups of uplift algorithms: Meta Learner (S-Learner, T-Learner, X-Learner) and Causal Forest, with a total of 39 algorithms available.

Uplift Modeling Algorithms



In this case study, we are using T-Learner Random Forest, T-Learner Logistic Regression, T-Learner Decision Tree, and Causal Forest algorithms.

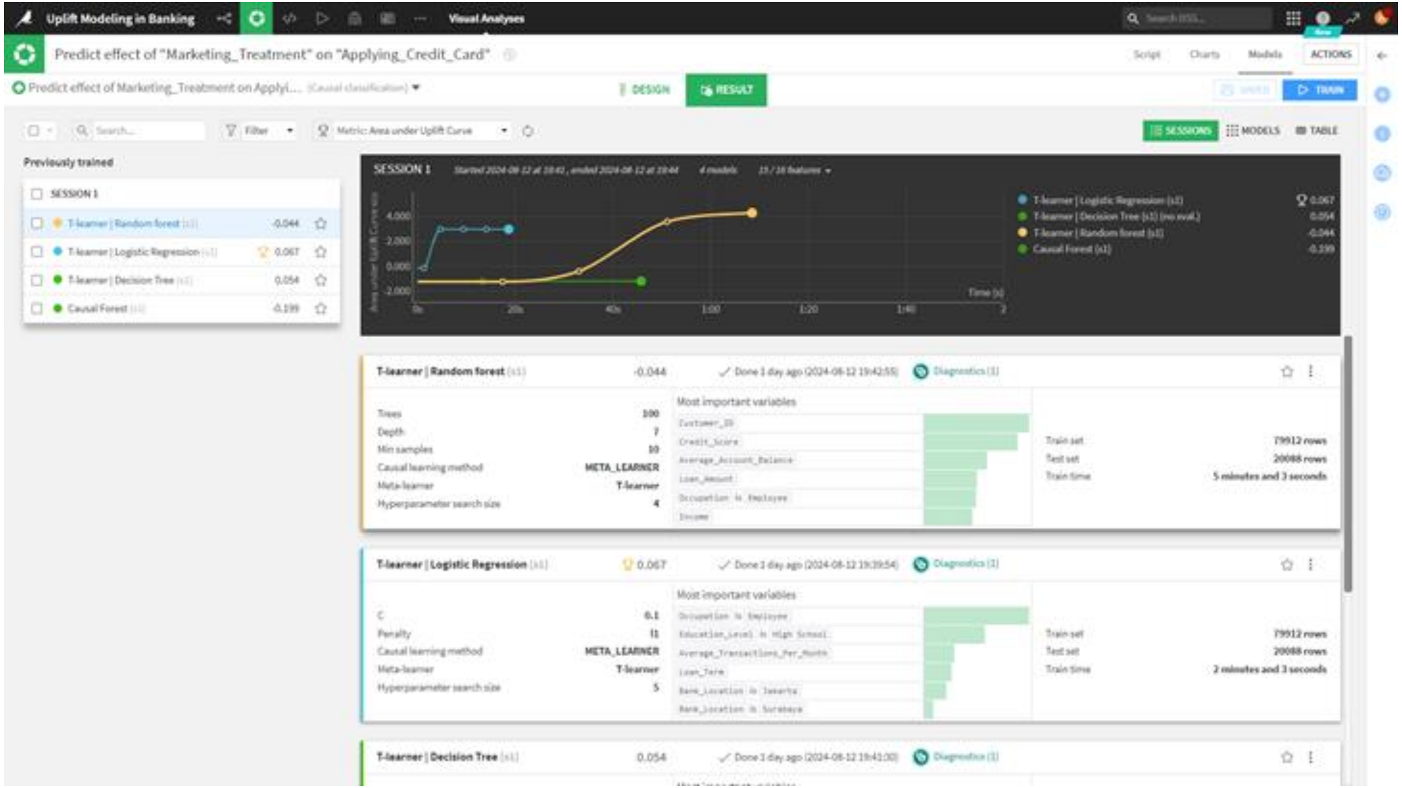
Uplift Modeling Algorithms



In this case study, we are using T-Learner Random Forest, T-Learner Logistic Regression, T-Learner Decision Tree, and Causal Forest algorithms.

Uplift Modeling Results

Uplift Training Results



T Learner | Logistic Regression" indicates that Logistic Regression is considered the best machine learning model for this particular analysis. This model is likely chosen for its effectiveness in predicting binary outcomes, such as whether a customer will apply for a credit card based on the marketing treatment they receive.

Detail T-Learner | Logistic Regression

T-learner | Logistic Regression (s1)

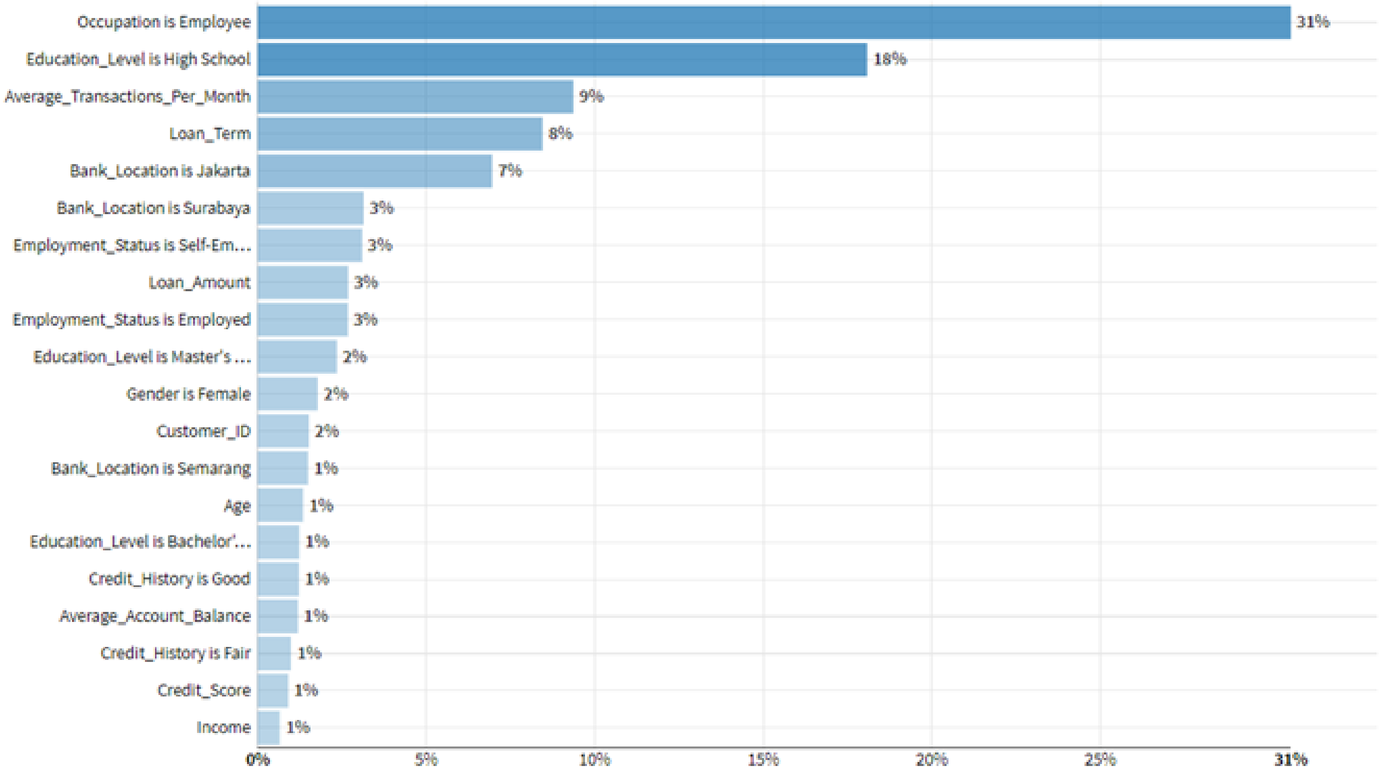
Area under Uplift Curve: 0.067

Model

Model ID	A-BANKING_UPLIFT-jWRbWVpo-T2wQWCMc-s1-pp2...
Model type	Causal classification
Outcome	Applying_Credit_Card
Treatment	Marketing_Treatment
Control value	0
Classes	0 1 (preferred class)
Backend	Python (in memory)
Algorithm	T-learner | Logistic regression
Trained on	2024/08/12 19:37
Columns	16
Train set rows	79912
Test set rows	20088
Code Env	new_python
Python version	3.9.2

Uplift Modeling Evaluation

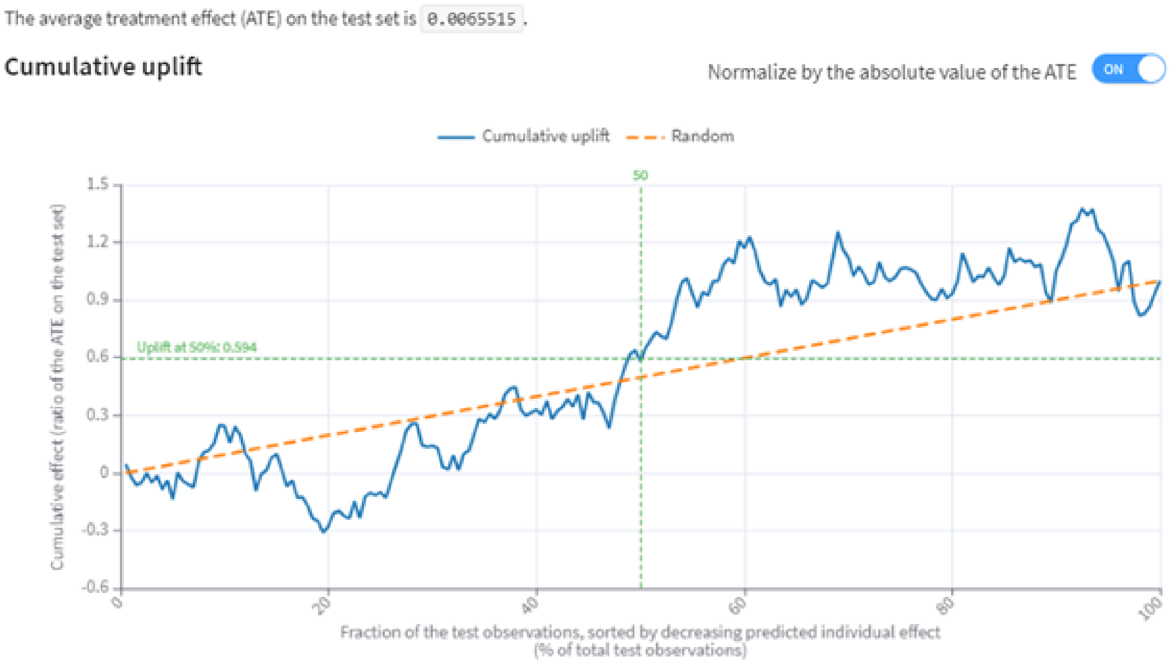
Feature Importance



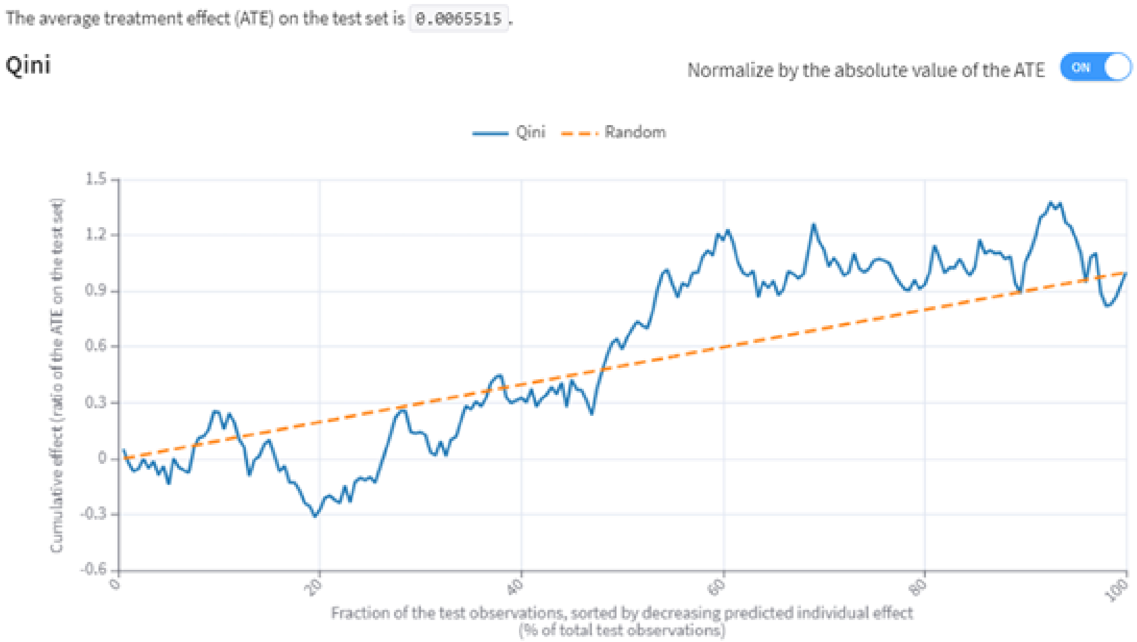
The image shows a horizontal bar graph highlighting factors that determine whether a customer is an employee. The most significant factor, "Occupation is Employee," contributes about 31%. Other key factors include "Average_Transaction_Per_Month," "Loan_Amount," and various attributes like employment status, gender, age, account balance, credit history, and income.

Cumulative Uplift

The cumulative uplift curve measures the cumulative uplift across subsets of the test set, scaled by the number of observations. A steeper curve near the origin indicates better performance. The baseline represents expected effects from random assignment, rising as the average treatment effect is positive. The AUUC (Area Under the Uplift Curve) is 0.066790, and the net uplift at 50% is 0.093533. These metrics assume randomized treatment, and may not be reliable if the treatment is not randomized.



Qini Curve



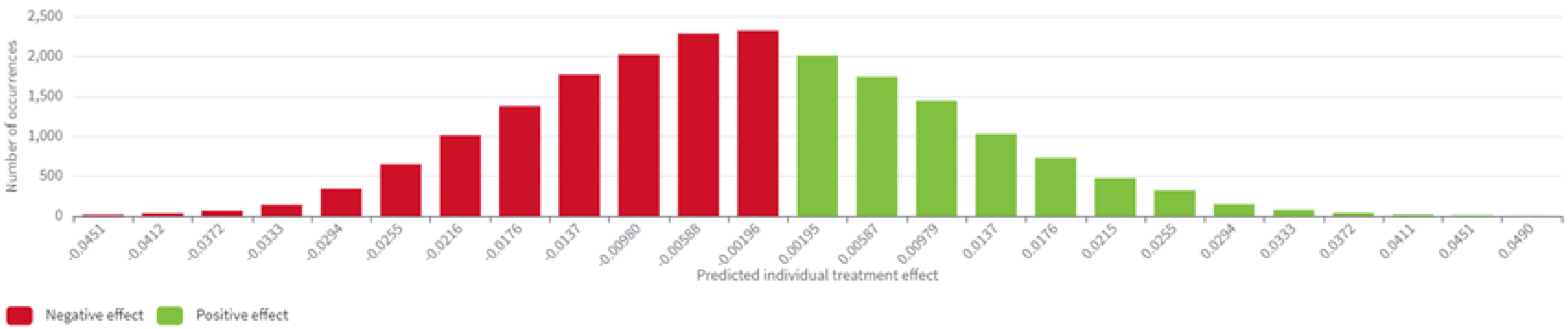
The Qini curve measures the cumulative uplift across subsets of the test set, scaled by the number of treated observations, with a steeper curve near the origin indicating better performance. The baseline shows expected effects from random assignment, rising with a positive average treatment effect. The Qini score, representing the area between the Qini curve and the random assignment line, is 0.067815. This metric assumes randomized treatment and may be unreliable if the treatment is not randomized.

Individual Treatment Effect

Predicted individual treatment effect distribution

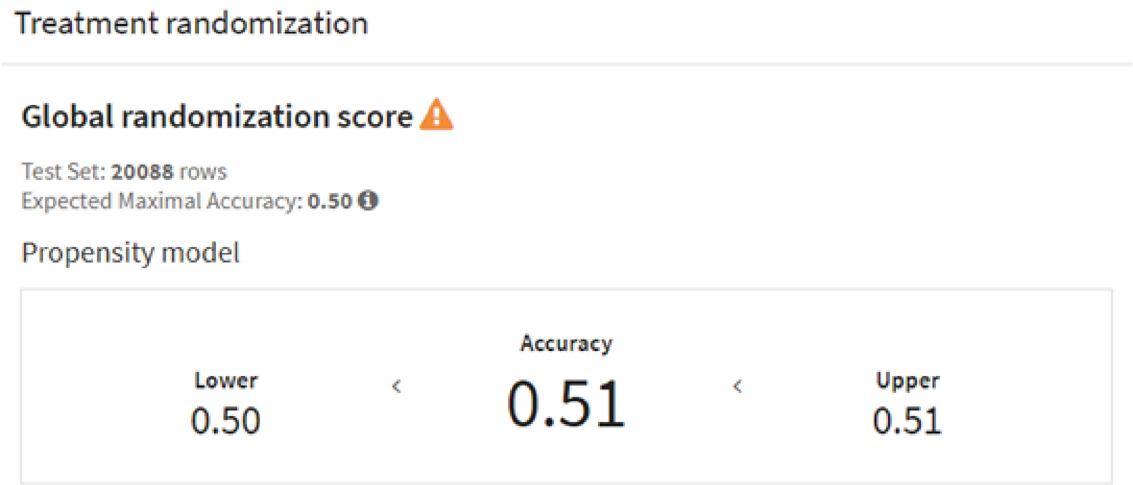
Description

This histogram shows the distribution of the **predicted individual treatment effect** over the test set (20088 observations).



This graph visualizes the distribution of predicted individual treatment effects over the last observations, indicating how many individuals are likely to experience positive or negative outcomes from a treatment. Description This histogram shows the distribution of the predicted individual treatment effect over the test set (20088 observations).

Global Randomization Score



Propensity Model Purpose:

- Discriminates between treated and control groups.
- Key Indicator: Accuracy > 0.50 suggests treatment is not randomized.

Binomial Test

Binomial test

Hypothesis tested	Treatment randomized (accuracy <= 0.50)
Significance level	0.050
p-value	0.047243
Conclusion	Treatment non-randomization detected

Binomial Test for Randomization

- Hypothesis:
  - Treatment is randomized, expecting model accuracy to be 0.50.

- Test Objective:
  - Evaluate if observed accuracy significantly deviates from 0.50.

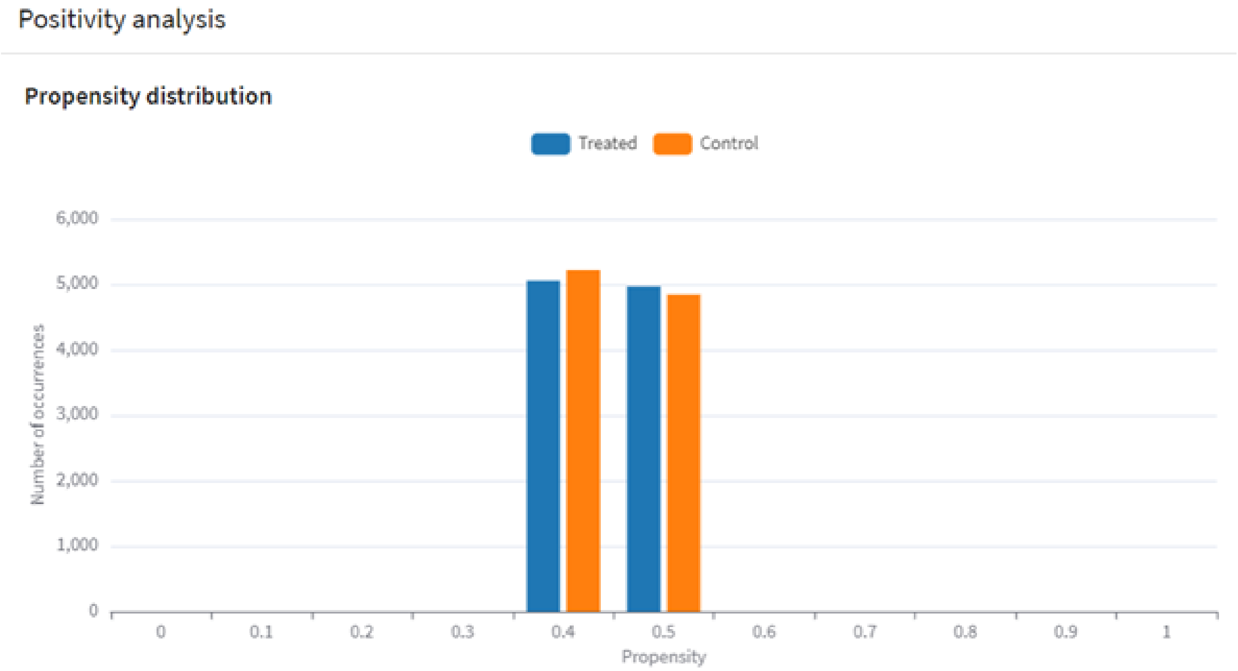
- P-Value Interpretation:
  - Low p-value (< 5%): Rejects randomization hypothesis, indicating non-randomized treatment.

- Significance Level:
  - Defines acceptable rate of false positives (detecting non-randomization when treatment is randomized).

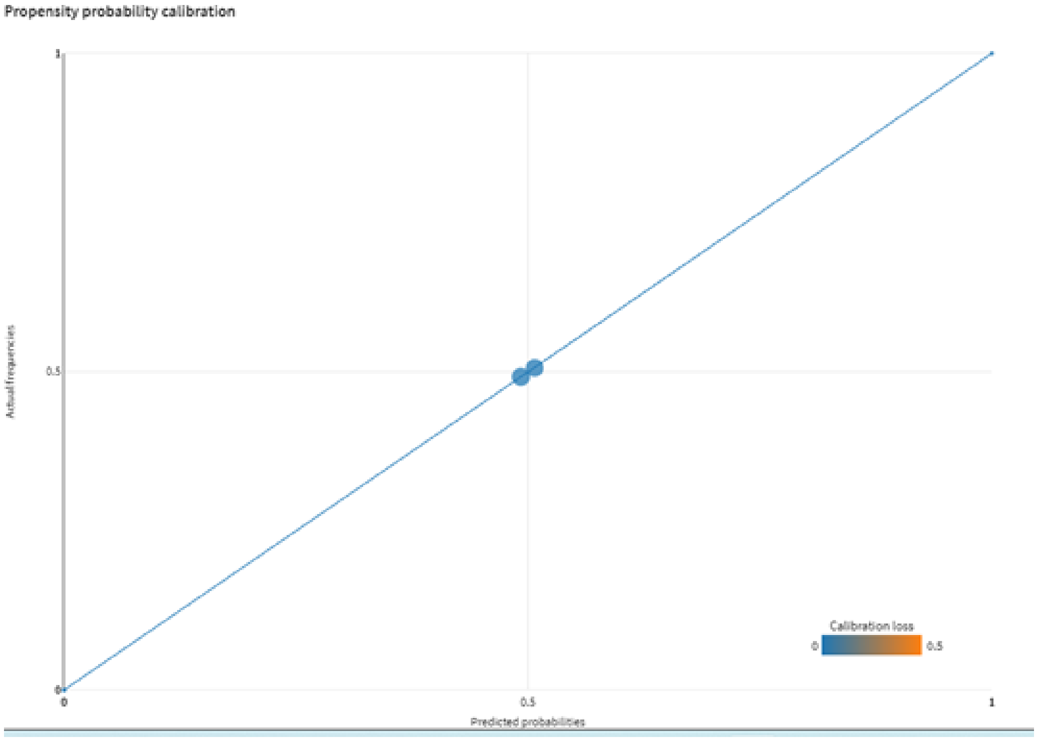
Propensity Positivity

Propensity positivity
The positivity assumption states that no subset of the feature space should have a null probability of treatment or control.

This chart can help detect if this assumption is violated. One way of finding such violations is to check that all bins of the probability distribution are not exclusively populated (in a statistically significant way) with either treated or control individuals.



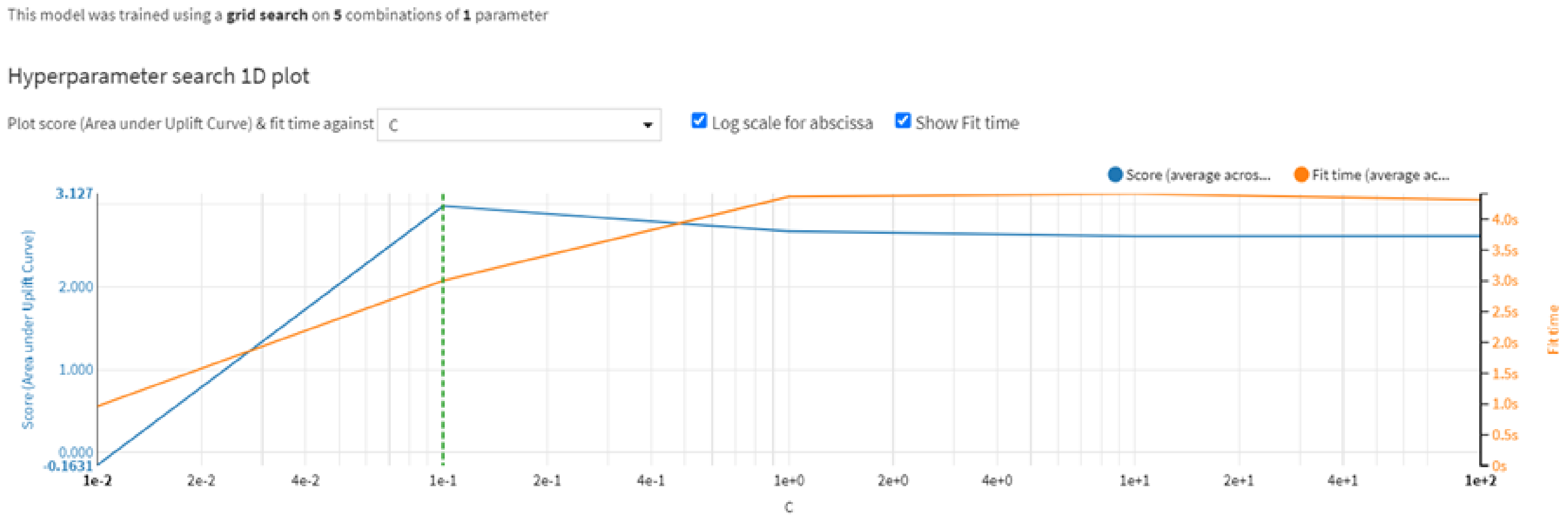
Calibration of predicted treatment probabilities



Calibration measures how well predicted probabilities align with actual outcomes on a test dataset. Calibration data points are calculated by averaging predictions and comparing them to observed frequencies within probability ranges (e.g., [0, 0.1), [0.1, 0.2), etc.).

A perfectly calibrated model's data points would align with the diagonal line, but deviations occur, measured as calibration loss. The calibration loss, averaging the absolute differences from the diagonal, is 0.0007. A calibration curve, a smoothed version of these data points, is also plotted against the diagonal line.

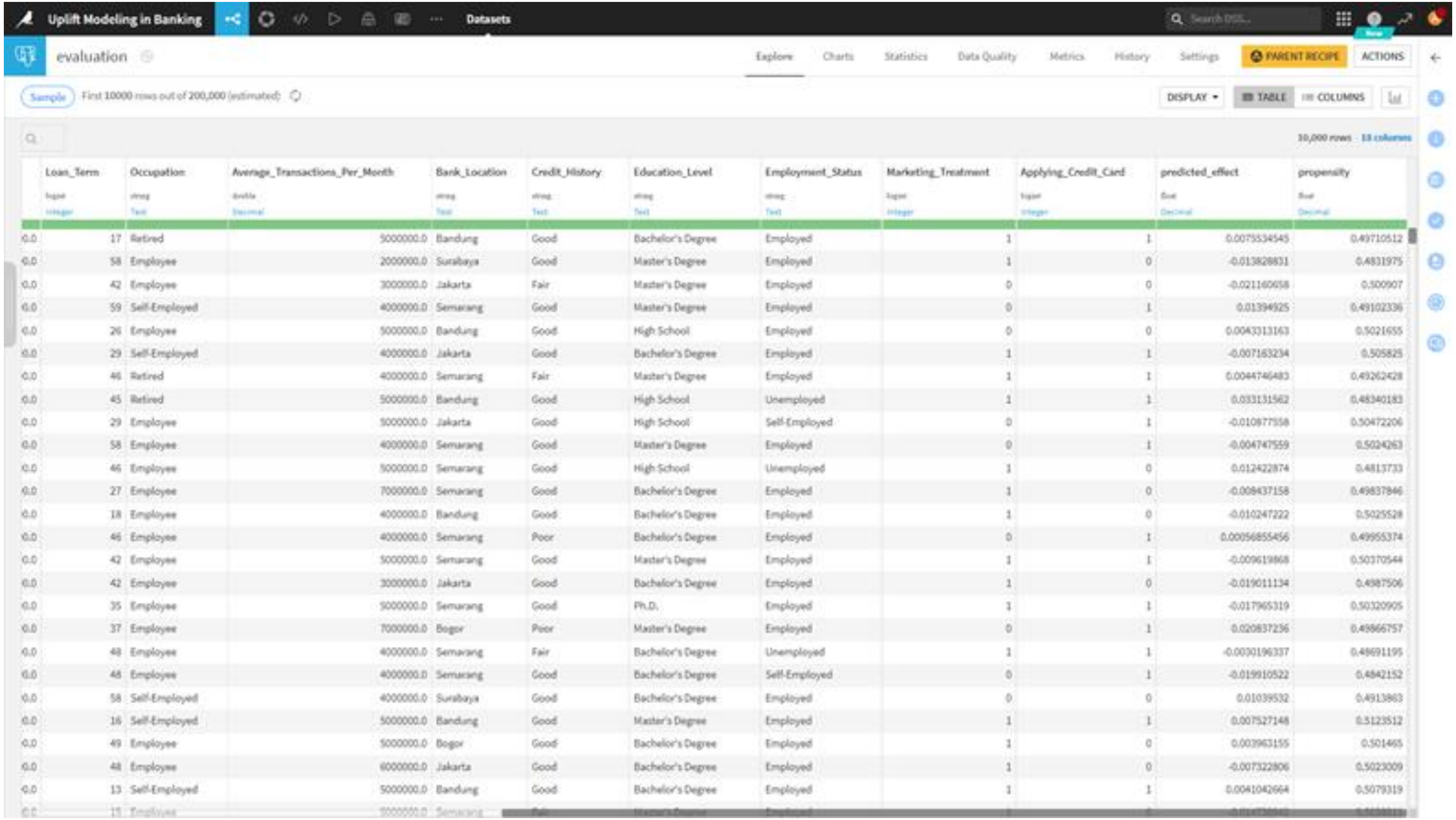
Hyperparameter Optimization



It shows the performance of a model with different hyperparameter combinations, focusing on the first score versus hand-tuned L1 ratio (first argument, C). This graph helps evaluate how different hyperparameter settings affect model performance and efficiency in machine learning tasks.

- Lines: Two lines represent “Score average across folds” in blue and “Runtime average across folds” in orange.
- Axes: The x-axis shows L1 ratio values from 0 to 1. The y-axis on the left represents score values, while the right side represents runtime in seconds.
- Trends: As the L1 ratio increases, the score initially rises sharply and then plateaus. Runtime remains relatively constant.

Data Evaluation

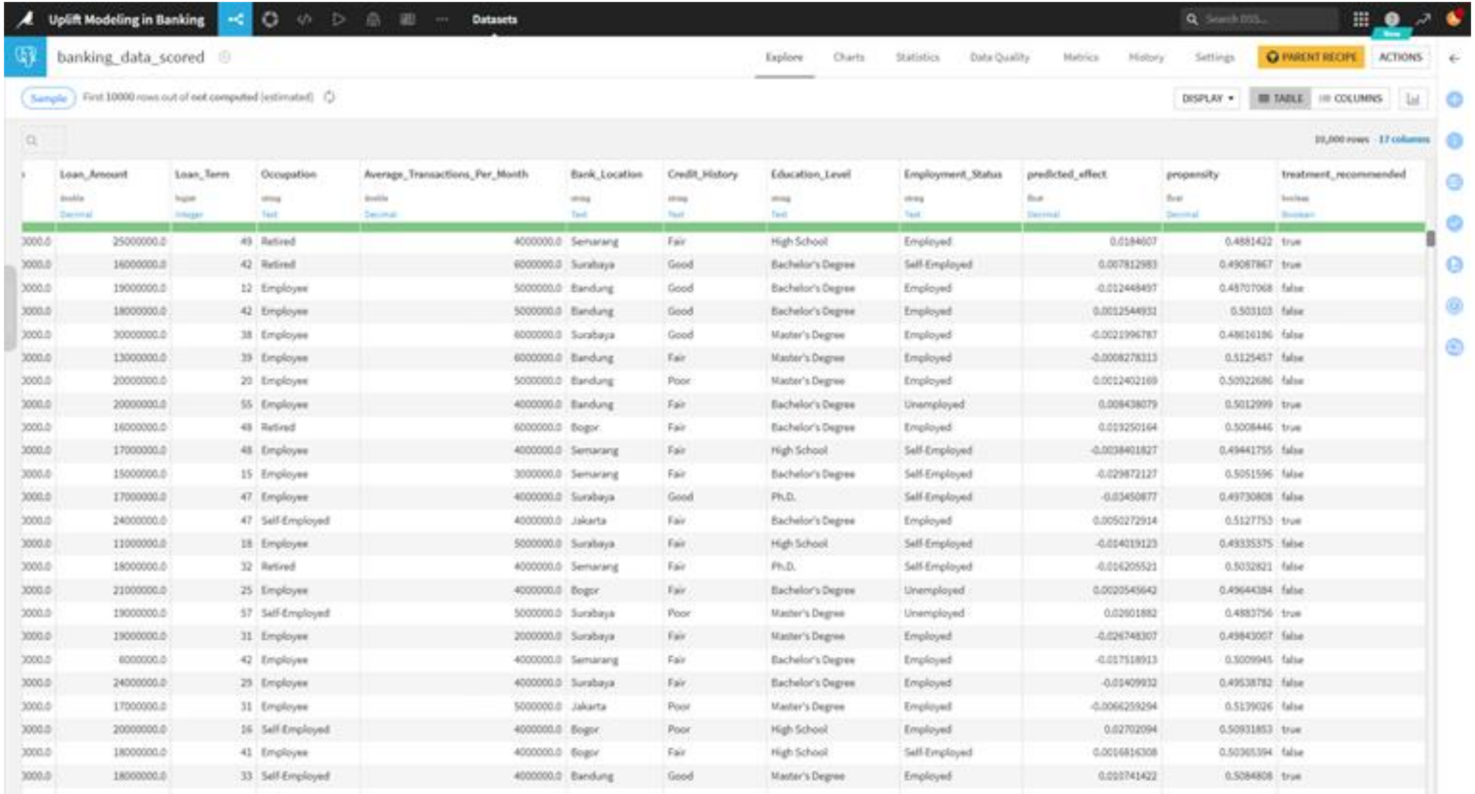


	Loan_Term	Occupation	Average_Transactions_Per_Month	Bank_Location	Credit_History	Education_Level	Employment_Status	Marketing_Treatment	Applying_Credit_Card	predicted_effect	propensity
0.0	17	Retired	5000000.0	Bandung	Good	Bachelor's Degree	Employed	1	1	0.0075534545	0.49710512
0.0	58	Employee	2000000.0	Surabaya	Good	Master's Degree	Employed	1	0	-0.013828831	0.4831975
0.0	42	Employee	3000000.0	Jakarta	Fair	Master's Degree	Employed	0	0	-0.021160658	0.500907
0.0	59	Self-Employed	4000000.0	Semarang	Good	Master's Degree	Employed	0	1	0.01394925	0.49102336
0.0	26	Employee	5000000.0	Bandung	Good	High School	Employed	0	0	0.0043313163	0.5021655
0.0	29	Self-Employed	4000000.0	Jakarta	Good	Bachelor's Degree	Employed	1	1	-0.007163234	0.505825
0.0	46	Retired	4000000.0	Semarang	Fair	Master's Degree	Employed	1	1	0.0044746483	0.49262428
0.0	45	Retired	5000000.0	Bandung	Good	High School	Unemployed	1	1	0.033131562	0.48340183
0.0	29	Employee	5000000.0	Jakarta	Good	High School	Self-Employed	0	1	-0.010877558	0.50472206
0.0	58	Employee	4000000.0	Semarang	Good	Master's Degree	Employed	0	1	-0.004747559	0.5024263
0.0	46	Employee	5000000.0	Semarang	Good	High School	Unemployed	1	0	0.012422874	0.4813733
0.0	27	Employee	7000000.0	Semarang	Good	Bachelor's Degree	Employed	1	0	-0.008437158	0.49837846
0.0	18	Employee	4000000.0	Bandung	Good	Bachelor's Degree	Employed	1	0	-0.010247222	0.5025528
0.0	46	Employee	4000000.0	Semarang	Poor	Bachelor's Degree	Employed	0	1	0.00056855456	0.49955374
0.0	42	Employee	5000000.0	Semarang	Good	Master's Degree	Employed	1	1	-0.009619868	0.50370544
0.0	42	Employee	3000000.0	Jakarta	Good	Bachelor's Degree	Employed	1	0	-0.019011134	0.4987506
0.0	35	Employee	5000000.0	Semarang	Good	Ph.D.	Employed	1	1	-0.017965319	0.50320905
0.0	37	Employee	7000000.0	Bogor	Poor	Master's Degree	Employed	0	1	0.020837236	0.49866757
0.0	48	Employee	4000000.0	Semarang	Fair	Bachelor's Degree	Unemployed	1	1	-0.0030196337	0.48691195
0.0	48	Employee	4000000.0	Semarang	Good	Bachelor's Degree	Self-Employed	0	1	-0.019910522	0.4842152
0.0	58	Self-Employed	4000000.0	Surabaya	Good	Bachelor's Degree	Employed	0	0	0.01039532	0.4913863
0.0	16	Self-Employed	5000000.0	Bandung	Good	Master's Degree	Employed	1	1	0.007527148	0.5123512
0.0	49	Employee	5000000.0	Bogor	Good	Bachelor's Degree	Employed	1	0	0.003963155	0.501465
0.0	48	Employee	6000000.0	Jakarta	Good	Bachelor's Degree	Employed	1	0	-0.007322806	0.5023009
0.0	13	Self-Employed	5000000.0	Bandung	Good	Bachelor's Degree	Employed	1	1	0.0041042664	0.5079319

Evaluation: Assess model predictions for accuracy using cross-validation and performance metrics.

Uplift Modeling Scoring

Uplift Scoring to Actual Data



If the treatment recommendation is true, then a positive uplift triggers the sending of marketing offers via email, SMS, and other channels. Conversely, if the treatment recommendation is false, it indicates no uplift or a negative impact, resulting in no marketing offer being sent, which helps in reducing costs.

Conclusion

- Precise Measurement: Uplift modeling in Dataiku allows for accurate measurement of the incremental impact of different email campaigns.
- Streamlined Process: Dataiku facilitate seamless data preparation, model building, and analysis.
- Effective Comparison: Enables effective comparison between personalized and generic offers.
- Enhanced Marketing Strategies: Identifies which treatments drive higher response rates, leading to more targeted campaigns.
- Optimized Resource Allocation: Helps in optimizing resource allocation and improving decision-making.
- Maximized Campaign ROI: Leveraging Dataiku for uplift modeling ultimately boosts the effectiveness of credit card marketing efforts and increases overall campaign ROI.

Recommendations for Accurate Uplift Modeling

- **Use Properly Randomized Real-World Data:** Ensures accurate treatment effects.
- **Validate Assumptions:** Check particularly for the positivity assumption to avoid subsets with null probability of treatment/control.
- **Calibrate Predicted Treatment Probabilities Regularly:** Align them with actual outcomes.
- **Optimize Hyperparameters:** Refine model performance for better results.
- **Evaluate Randomization:** Use metrics like the Global Randomization Score and Binomial Test.
- **Leverage Dataiku:** Utilize these for data preparation, model building, and analysis to enhance uplift modeling efficiency.

Reference

https://www.dataiku.com/learn/samples/uplift-modelling/

https://www.youtube.com/watch?v=fInFcWcmOeU&pp=ygUQdXBsaWZ0IG1vZGVsbGluZw%3D%3D

Thank You